



### ASSIGNMENT

| Marks      |          | Course No.           | EEE 4167          |
|------------|----------|----------------------|-------------------|
| Allocated  | Obtained | Course Title         | Optoelectronics   |
| 10         |          | Course Instructor(s) | Md. Khayrul Islam |
|            |          | Due Date             | October 18, 2025  |
|            |          | Submission Date      |                   |
| Student ID |          | Student Name         |                   |

#### Problem Statement:

Design and analyze an optical modulation system for controlling or detecting light intensity, phase, or polarization.

#### Description of the Problem Statement:

Your task is to design a **modulation-based optical system** that demonstrates how light can be **controlled or detected** using one or more **modulation mechanisms** such as the **electro-optic effect, magneto-optic effect, or acousto-optic effect**. The aim is to analyze how these phenomena modify the properties of light (intensity, polarization, or phase) and how they are applied in modern optoelectronic communication or sensing systems.

#### General Instructions:

The solution of the assignment must contain:

1. **Problem Statement:**

Clearly describe the selected modulation technique and its practical importance in light control or detection.

2. **Proposed Model:**

- Present a **conceptual or schematic diagram** of your modulation-based optical system (clearly labeling source, modulator, and detector).
- Explain the **physical principle** (e.g., electro-optic, magneto-optic, or acousto-optic effect) governing modulation.
- Provide **relevant mathematical relations**, such as those for phase retardation, intensity variation, or polarization rotation.

3. **Results and Discussion:**

- Include **theoretical, graphical, or simulated results** showing the relationship between input field (electric, magnetic, or acoustic) and output optical intensity or phase.
- Discuss how modulation depth, bandwidth, and response time affect detection accuracy.
- Relate the modulation process to real-world applications (e.g., optical communication, sensors, or laser systems).



4. **Addressing Course Learning Outcome (CLO2) and Program Learning Outcome (PLO3):**  
Explain how your system demonstrates the design and evaluation of a light detection mechanism through modulation.
5. **Addressing Complex Engineering Problems (P1–P3):**
  - **P1:** Depth of theoretical and practical knowledge in optical modulation and detection.
  - **P2:** Trade-offs between modulation efficiency, response time, and material limitations.
  - **P3:** Depth of analysis required to model system behavior mathematically or experimentally.
6. **Conclusion:**  
Summarize your findings and discuss how light modulation enhances performance in optoelectronic detection systems.
7. **References:**  
Cite all books, papers, and online sources (IEEE referencing style).

**Note:** Plagiarized or AI-generated content will result in a zero mark.

#### Addressing CLO and PLO

| Attribute                                                                                                              | Justification                                                                                                                                                                      |
|------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CLO2:</b> Detect light by designing various optical detectors with the help of different photodetection mechanisms. | This assignment fulfills CLO2 by requiring students to design a system that detects or analyzes light through different modulation mechanisms, integrating theory and application. |
| <b>PLO3:</b> Design/Development of Solution                                                                            | The assignment involves the design and analysis of an optoelectronic modulation system, emphasizing engineering solutions for light control and detection.                         |

#### Addressing Complex Engineering Problems (P's):

| Attribute                                    | Justification                                                                                                                            |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <b>P1:</b> Depth of Knowledge Required       | Requires comprehensive understanding of electro-optic, magneto-optic, and acousto-optic physics, as well as optical material properties. |
| <b>P2:</b> Range of Conflicting Requirements | Balances modulation efficiency, optical losses, device size, and speed constraints.                                                      |
| <b>P3:</b> Depth of Analysis Required        | Involves mathematical modeling and evaluation of optical modulation and detection performance.                                           |



**Rubrics (10 Marks Total)**

| Source                      | Allocated Marks | Corresponding CLO | Distribution of Marks                                                                                                                                               | Obtained Marks |
|-----------------------------|-----------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| Proposed Model              | 4               | CLO2              | <b>4:</b> Clear, detailed schematic and theoretical explanation.<br><b>2–3:</b> Adequate but lacks depth or labeling.<br><b>0–1:</b> Incomplete or incorrect model. |                |
| Results and Discussion      | 3               | CLO2              | <b>3:</b> Strong analytical or simulated discussion.<br><b>1–2:</b> Basic or partially analyzed results.<br><b>0:</b> Missing or incorrect discussion.              |                |
| Addressing CLO and PLO      | 1               | CLO2              | <b>1:</b> Clearly justified.<br><b>0:</b> Not addressed.                                                                                                            |                |
| Addressing P's (P1–P3)      | 1               | CLO2              | <b>1:</b> Clearly justified.<br><b>0:</b> Not addressed.                                                                                                            |                |
| Conclusion                  | 1               | CLO2              | <b>1:</b> Clear, concise, and relevant conclusion.<br><b>0:</b> Generic or missing.                                                                                 |                |
| <b>Total Obtained Marks</b> |                 |                   |                                                                                                                                                                     |                |



### Annexure

#### Mapping of Course Outcomes (COs), Program Outcomes (POs), Teaching-Learning Strategies, and Assessment Tools:

| CLO No. | CLO Statement                                                                                             | Corresponding PLO                     | Domain & *level of learning taxonomy | Delivery methods and activities                                                                                                                                                                                                                                                          | Assessment tools                                                                                                                                                                                                                                                         |
|---------|-----------------------------------------------------------------------------------------------------------|---------------------------------------|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CLO1    | Analyze the performance of modulators, LEDs, LASERs, various detectors and integrated waveguides          | PLO2: Problem Analysis                | Cognitive (Analyze)                  | <input checked="" type="checkbox"/> Lecture<br><input type="checkbox"/> Tutorial<br><input checked="" type="checkbox"/> Discussion<br><input type="checkbox"/> Interaction<br><input checked="" type="checkbox"/> Audio/Video<br><input type="checkbox"/> Others: If any please mention. | <input checked="" type="checkbox"/> Class Test<br><input type="checkbox"/> Assignment<br><input checked="" type="checkbox"/> Final Exam<br><input type="checkbox"/> Presentation<br><input type="checkbox"/> Project<br><input type="checkbox"/> Others: Quiz            |
| CLO2    | Detect light by designing various optical detectors with the help of different photodetection mechanisms. | PLO3: Design/ Development of Solution | Cognitive (Evaluate)                 | <input checked="" type="checkbox"/> Lecture<br><input type="checkbox"/> Tutorial<br><input type="checkbox"/> Discussion<br><input type="checkbox"/> Interaction<br><input checked="" type="checkbox"/> Audio/Video<br><input type="checkbox"/> Others: If any please mention.            | <input checked="" type="checkbox"/> Class Test<br><input checked="" type="checkbox"/> Assignment<br><input checked="" type="checkbox"/> Final Exam<br><input type="checkbox"/> Presentation<br><input type="checkbox"/> Project<br><input type="checkbox"/> Others: Quiz |

\*Levels in Bloom's Cognitive Domain: Level 1: Remember, Level 2: Understand, Level 3: Apply, Level 4: Analyze, Level 5: Evaluate, Level 6: Create

\*Levels in Bloom's Psychomotor Domain: Level 1: Imitate, Level 2: Manipulate, Level 3: Precise, Level 4: Articulate, Level 5: Naturalize

\*Levels in Bloom's Affective Domain: Level 1: Receive, Level 2: Respond, Level 3: Value, Level 4: Organize, Level 5: Characterize



### Mapping of Knowledge Profile, Complex Engineering Problems and Complex Engineering Activities:

| Program Learning outcome (PLO) |                       |      |                  |      |                    |      |               |      |                   |      |                    | Knowledge Profile |                                |      |        | Complex Engineering Problem |                              |       |               | Complex Engineering Activities |                                |       |                    |           |  |  |  |      |  |  |  |      |  |  |  |           |  |  |  |      |  |  |  |           |  |  |  |       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
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| PLO1                           | Engineering Knowledge | PLO2 | Problem Analysis | PLO3 | Design of Solution | PLO4 | Investigation | PLO5 | Modern Tool Usage | PLO6 | Engineer & Society | PLO7              | Environment and sustainability | PLO8 | Ethics | PLO9                        | Individual work and teamwork | PLO10 | Communication | PLO11                          | Project management and finance | PLO12 | Life-long learning | PLO1-PLO2 |  |  |  | PLO3 |  |  |  | PLO5 |  |  |  | PLO6-PLO8 |  |  |  | PLO4 |  |  |  | PLO1-PLO7 |  |  |  | PLO10 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| ✓                              |                       |      |                  |      |                    |      |               |      |                   |      |                    |                   |                                |      |        |                             |                              |       |               |                                |                                |       |                    |           |  |  |  |      |  |  |  |      |  |  |  |           |  |  |  |      |  |  |  |           |  |  |  |       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |

**Table 1: Knowledge Profile**

| Attribute |                                                                                                                                                                                                                                                                                                |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| K1        | A systematic, theory-based understanding of the natural sciences applicable to the discipline                                                                                                                                                                                                  |
| K2        | Conceptually based mathematics, numerical analysis, statistics and the formal aspects of computer and information science to support analysis and modeling applicable to the discipline                                                                                                        |
| K3        | A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline                                                                                                                                                                                      |
| K4        | Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline                                                                                        |
| K5        | Knowledge that supports engineering design in a practice area                                                                                                                                                                                                                                  |
| K6        | Knowledge of engineering practice (technology) in the practice areas in the engineering discipline                                                                                                                                                                                             |
| K7        | Comprehension of the role of engineering in society and identified issues in engineering practice in the discipline: ethics and the engineer's professional responsibility to public safety; the impacts of engineering activity; economic, social, cultural, environmental and sustainability |
| K8        | Engagement with selected knowledge in the research literature of the discipline                                                                                                                                                                                                                |



**Table 2: Range of Complex Engineering Problem Solving**

| Attribute                                                      |                                                                                                                                                                                                  |
|----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Depth of knowledge required                                    | <b>P1:</b> Cannot be resolved without in-depth engineering knowledge at the level of one or more of K3, K4, K5, K6 or K8 which allows a fundamentals-based, first principles analytical approach |
| Range of conflicting requirements                              | <b>P2:</b> Involve wide-ranging or conflicting technical, engineering and other issues                                                                                                           |
| Depth of analysis required                                     | <b>P3:</b> Have no obvious solution and require abstract thinking, originality in analysis to formulate suitable models                                                                          |
| Familiarity of issues                                          | <b>P4:</b> Involve infrequently encountered issues                                                                                                                                               |
| Extent of applicable codes                                     | <b>P5:</b> Are outside problems encompassed by standards and codes of practice for professional engineering                                                                                      |
| Extent of stakeholder involvement and conflicting requirements | <b>P6:</b> Involve diverse groups of stakeholders with widely varying needs                                                                                                                      |
| Interdependence                                                | <b>P7:</b> Are high-level problems including many component parts or sub-problems                                                                                                                |

**Table 3: Range of Complex Engineering Activities**

| Attribute                                    |                                                                                                                                                            |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Range of resources                           | <b>A1:</b> Involve the use of diverse resources (and for this purpose resources include people, money, equipment, materials, information and technologies) |
| Level of interaction                         | <b>A2:</b> Require resolution of significant problems arising from interactions between wide-ranging or conflicting technical, engineering or other issues |
| Innovation                                   | <b>A3:</b> Involve creative use of engineering principles and research-based knowledge in novel ways                                                       |
| Consequences for society and the environment | <b>A4:</b> Have significant consequences in a range of contexts, characterized by difficulty of prediction and mitigation                                  |
| Familiarity                                  | <b>A5:</b> Can extend beyond previous experiences by applying principles-based approaches                                                                  |